

10.10 We have to account for models denoted by m , and since we are not considering a factorized approximation $q(\mathbf{Z}) q(m)$, we start with their joint distributions:

$$\begin{aligned}\mathcal{L}(q) &= \sum_m \sum_{\mathbf{Z}} q(\mathbf{Z}, m) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}, m)}{q(\mathbf{Z}, m)} \right\} \\ &= \sum_m \sum_{\mathbf{Z}} q(\mathbf{Z}|m) q(m) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}, m)}{q(\mathbf{Z}|m) q(m)} \right\} \\ \\ KL(q||p) &= - \sum_m \sum_{\mathbf{Z}} q(\mathbf{Z}, m) \ln \left\{ \frac{p(\mathbf{Z}, m|\mathbf{X})}{q(\mathbf{Z}, m)} \right\} \\ &= - \sum_m \sum_{\mathbf{Z}} q(\mathbf{Z}|m) q(m) \ln \left\{ \frac{p(\mathbf{Z}, m|\mathbf{X})}{q(\mathbf{Z}|m) q(m)} \right\}\end{aligned}$$

Combining the two, we get:

$$\ln p(\mathbf{X}) = \mathcal{L}(q) - \sum_m \sum_{\mathbf{Z}} q(\mathbf{Z}|m) q(m) \ln \left\{ \frac{p(\mathbf{Z}, m|\mathbf{X})}{q(\mathbf{Z}|m) q(m)} \right\}$$

which is the same as (10.34).